

On the Universality of Comparability Grids for Measurement-Based Quantum Computation

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Abstract

TBD

Acknowledgements

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Preface to the Rough Draft

One of the great difficulties writing the rough draft was figuring out where to stop. In other words, how much can I assume from the reader? To answer this question, the past honors theses of Ray Tsai, Clara Woods, and Jared Hughes in particular have been most helpful; the first for its recency and treatment of graph theory, the second for its experimental and quantum focus, and the third for its approach to 3XOR and theoretical computer science. These three, among other theses, appear to take a very introductory stance, writing for a non-expert audience familiar with only the typical mathematical fields (i.e. a senior mathematics undergraduate).

Because of this, I have prepared a (perhaps excessively) extensive preliminary section. I was originally planning to cut out the preliminary section entirely, or perhaps place it some extended appendix. However, no prior honors thesis (to the best of my skimming) has ever used the appendix in such a manner, nor excluded so much introduction. Thus, since it is easier to cut than to add, I have left it all in.

Furthermore, the main results of the research connected to this thesis is still a work in progress. Because of this, I found it challenging to create some sort of cohesive presentation of the work besides the chronological events of discoveries. Because of this, I recognize much work needs to be done to better distinguish the original research from what is known in the literature.

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1 Introduction

We start with an informal introduction to quantum computing and lay some motivations.

Fault-tolerant quantum computing has the potential, through algorithms like Shor’s factoring algorithm [11], to deliver possibly exponential speedups over classical computation. However, current Noisy Intermediate-Scale Quantum (NISQ) quantum computers have too few qubits and too much noise to deliver significant practical benefit [9].

Computation in the NISQ era is often modelled by a hybrid setup: on the quantum computer, a noisy initial state is prepared, some noisy gates are applied, and qubits are measured. A classical computer then processes these measurements, and can modify and repeat the quantum computation if necessary.

For the quantum computer, there exist a variety of computational models: The quantum Turing machine [5], the quantum circuit model [2], adiabatic quantum computation [6], measurement-based models [10], among many others. Though these models are equivalent from a complexity viewpoint (informally, any problem that is efficiently solvable in one is efficiently solvable in the others) [10, 1, 4], the underlying quantum properties used by each may differ significantly.

The measurement-based quantum computation (MBQC) model is of particular interest. This is because MBQC collapses any quantum circuit into measurements of a highly-entangled resource state that can be prepared in constant depth. This is quite shocking, as no clear classical counterpart exists. The cost of this model, however, is that measurements need to be adaptive. In other words, some post-processing must be done on-the-fly to determine what qubits should be measured next.

Since measurement is irreversible, MBQC leads to a model of one-way quantum computation. In other words, as measurement cannot increase the entanglement of a system, all entanglement required must be initially present in the resource state. Informally, entanglement is the resource consumed by this model of quantum computation. Thus, there is significant interest in determining what resource states are universal for quantum computation. That is, what resource states, under the MBQC model, can generate any possible arrangement of qubits?

The comparability grid is the graph with vertices $V = \{(x, y) \mid x, y \in [n]\}$ and an edge from (x_1, y_1) to (x_2, y_2) if $x_1 \geq x_2$ and $y_1 \geq y_2$. It is an open problem whether the family of all comparability grids is universal for MBQC. There are powerful theoretical implications for both possibilities. For example, if this family is universal, then any universal family of graph states must have unbounded rank width [cite:KunalMD]. If comparability grids are not universal, then unbounded rank width is not a sufficient condition, since the family of all comparability grids have unbounded rank width.

In this thesis, we explore the comparability grid’s potential as a resource state for universal quantum computation. In section 2, we introduce the preliminaries necessary for such an endeavour, before exploring various algorithmic and theoretical approaches in later sections. We conclude with a discussion of similar results pertaining to circle graphs.

2 Preliminaries

First some notation. Let $\mathbb{N} = \{1, 2, \dots\}$. For $k \in \mathbb{Z}_{\geq 0}$, define $[k] := \{1, \dots, k\}$. Let I be the identity matrix of any size and of any vector space, which will be clear from context. For a graph G , let $V(G)$ denote its vertex set and $E(G)$ its edge set. For any vector v with n entries, let v_i be its i th entry for all $i \in [n]$.

2.1 Qubits

DEFINITION 2.1 A single **qubit** $|\phi\rangle$ is a state

$$|\phi\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{C}^2, \quad (1)$$

where $|\alpha|^2 + |\beta|^2 = 1$ and $\alpha, \beta \in \mathbb{C}$. We say $|\phi\rangle$ is in a **superposition** between $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Thus a single qubit is a length 2 vector in the Hilbert space $\mathbb{C}^2$ with a ℓ_2 norm of 1.

DEFINITION 2.2 Define kets $|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, both in $\mathbb{C}^2$. For any $|\phi\rangle$, let bra $\langle\phi| := |\phi\rangle^\dagger = (|\phi\rangle^*)^T$ be its conjugate transpose. Thus the bracket denotes the inner product:

$$\langle\phi|\phi\rangle := \langle\phi| \cdot |\phi\rangle. \quad (2)$$

Also, the ℓ_2 norm or length $||\phi\rangle| := \sqrt{\langle\phi|\phi\rangle}$. In general, for any $|\phi\rangle$ and $|\psi\rangle$,

$$\langle\phi|\psi\rangle := \langle\phi| \cdot |\psi\rangle = \sum_{i=1}^2 \phi_i^* \psi_i. \quad (3)$$

DEFINITION 2.3 Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{o \times p}$. For our purposes, it suffices to define the **tensor product**

$$A \otimes B := \begin{pmatrix} a_{1,1}B & \cdots & a_{1,n}B \\ \vdots & \ddots & \vdots \\ a_{n,1}B & \cdots & a_{n,n}B \end{pmatrix} \quad (4)$$

where $a_{i,j}$ denotes the i th row and j th column of A . If $x = \overline{x_1 x_2 \cdots x_d} \in \{0, 1\}^d$,

$$|x\rangle := \bigotimes_{i=1}^d |x_i\rangle. \quad (5)$$

We may replace x by its integer evaluation as a binary number. Also, we may write $|\phi\rangle |\psi\rangle = |\phi\rangle \otimes |\psi\rangle$, so adjacent kets imply $\otimes$. Also, let $A^{\otimes n} := \bigotimes_{i=1}^n A$. This gives us the definition of a d -qubit state

$$|\varphi\rangle = \sum_{x \in \{0,1\}^d} \alpha_x |x\rangle \in \mathbb{C}^{2^d}, \quad (6)$$

where $\alpha_x \in \mathbb{C}$ and $\sum_{x \in \{0,1\}^d} \alpha_x |x\rangle = 1$. As the number of dimensions increases exponentially with regards to the number of qubits, bracket notation will become increasingly concise compared to traditional vector notation.

DEFINITION 2.4 Vectors $|v_1\rangle, \dots, |v_n\rangle \in \mathbb{C}^n$ are an orthonormal basis if

1. $\| |v_i\rangle \| = 1$ for every $i \in [n]$ and
2. $|v_i\rangle \perp |v_j\rangle$ (equivalently, $\langle v_i | v_j \rangle = 0$) for every $i, j \in [n]$ with $i \neq j$.

Henceforth, all bases are orthonormal unless explicitly said otherwise.

DEFINITION 2.5 The **Born Rule** defines how quantum measurement occurs: A n -qubit state $|\phi\rangle$ can be measured under any basis $|v_i\rangle \in \mathbb{C}^{2^n}$ with $i \in [2^n]$. When measured, the output i (interpreted as classical bits) occurs with probability $|\langle \phi | v_i \rangle|^2$. After seeing the outcome v_i , the state collapses to $|v_i\rangle$. This process is irreversible.

DEFINITION 2.6 One can also view measurement as projection. This is useful for **partial measurement**, where only some qubits in a state are measured. Let the n qubit space $\mathbb{C}^{2^n} = S_1 \otimes S_2$ where $S_1 \cong (\mathbb{C}^2)^k$. Let $\{|s_i\rangle\}_{i \in [k]}$ be a basis for S_1 . Define the **projection matrix** on S_1 as

$$P := \sum_{j=1}^k |s_j\rangle \langle s_j|, \quad \text{such that} \quad P = P^\dagger, \quad P^2 = P. \quad (7)$$

Define the **partial measurement** on S_1 by $M = P \otimes I \in \mathbb{C}^{2^n \times 2^n}$. For a state $|\phi\rangle \in \mathbb{C}^{2^n}$,

$$\Pr(\text{Measuring } i \text{ from } |\phi\rangle) = \langle \phi | M | \phi \rangle = \|M | \phi \rangle\|^2. \quad (8)$$

On measuring outcome i , the post-measurement state becomes

$$|\phi_i\rangle = \frac{M | \phi \rangle}{\sqrt{\langle \phi | M | \phi \rangle}}. \quad (9)$$

In other words, $|\phi_i\rangle$ is the state that minimizes $\| |\phi\rangle - |\phi_j\rangle \|$ over all $|\phi_j\rangle \in S_1$. Partial measurements technically give rise to mixed states, which cannot be represented as a single ket, but we defer its discussion to a later section.

EXAMPLE 2.7 Consider the states

$$|+\rangle := \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle, \quad |-\rangle := \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle. \quad (10)$$

Measuring either in the **standard basis** $\{|0\rangle, |1\rangle\}$ gives $|0\rangle$ with 0.5 probability and $|1\rangle$ with 0.5 probability since

$$|\langle + | 0 \rangle|^2 = \left| \frac{1}{\sqrt{2}} \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 \right|^2 = \frac{1}{2}$$

and similarly for $|1\rangle$. The squaring makes $|-\rangle$ proceed similarly as well.

DEFINITION 2.8 A two qubit state $|\phi\rangle$ is **entangled** if it cannot be written as the tensor product of single qubits. That is, $\nexists |\psi\rangle, |\varphi\rangle \in \mathbb{C}^2$ such that $|\phi\rangle = |\psi\rangle \otimes |\varphi\rangle$. If a state is not entangled, it is called **separable**.

2.2 Quantum Gates

DEFINITION 2.9 A single qubit quantum gate $U \in \mathbb{C}^{2 \times 2}$ satisfies one of:

1. $U^\dagger U = U U^\dagger = I$;
2. equivalently, U is a unitary matrix;
3. equivalently, $U = |\phi_0\rangle\langle 0| + |\phi_1\rangle\langle 1|$, where $\{|\phi_0\rangle, |\phi_1\rangle\}$ form an orthonormal basis.

Common single qubit gates are given in Figure 1.

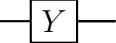
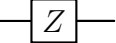
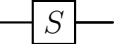
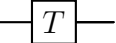
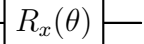
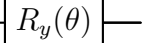
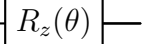
Name	Circuit	Matrix	Bra-Ket
X		$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$ 1\rangle\langle 0 + 0\rangle\langle 1 $
Y		$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	$i 1\rangle\langle 0 - i 0\rangle\langle 1 $
Z		$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$ 0\rangle\langle 0 - 1\rangle\langle 1 $
H		$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$ +\rangle\langle 0 + -\rangle\langle 1 $
S		$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$ 0\rangle\langle 0 + i 1\rangle\langle 1 $
T		$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	$ 0\rangle\langle 0 + e^{i\pi/4} 1\rangle\langle 1 $
$R_x(\theta)$		$\begin{pmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$	$\cos(\theta/2) 0\rangle\langle 0 - i \sin(\theta/2) 1\rangle\langle 0 $ $-i \sin(\theta/2) 0\rangle\langle 1 + \cos(\theta/2) 1\rangle\langle 1 $
$R_y(\theta)$		$\begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$	$\cos(\theta/2) 0\rangle\langle 0 + \sin(\theta/2) 1\rangle\langle 0 $ $-\sin(\theta/2) 0\rangle\langle 1 + \cos(\theta/2) 1\rangle\langle 1 $
$R_z(\theta)$		$\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$	$e^{-i\theta/2} 0\rangle\langle 0 + e^{i\theta/2} 1\rangle\langle 1 $

Figure 1: Common single qubit quantum gates

FACT 2.10 For matrices A, B, C, D , scalar λ , and vectors $|\phi\rangle, |\psi\rangle$, $\otimes$ satisfies:

1. Bi-linearity: $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, so $(A \otimes B)(|\phi\rangle \otimes |\psi\rangle) = (A|\phi\rangle) \otimes (B|\psi\rangle)$.
2. Scalars commute: $\lambda(A \otimes B) = (\lambda A) \otimes B = A \otimes (\lambda B)$.
3. Associativity: $(A \otimes B) \otimes C = A \otimes (B \otimes C)$.
4. Unitary-preserving: If A, B are unitary matrices, $A \otimes B$ is unitary.

DEFINITION 2.11 The general 1-qubit unitary is given by

$$U(\theta, \iota, \kappa) := \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -e^{i\kappa} \sin\left(\frac{\theta}{2}\right) \\ e^{i\iota} \sin\left(\frac{\theta}{2}\right) & e^{i(\iota+\kappa)} \cos\left(\frac{\theta}{2}\right) \end{pmatrix}. \quad (11)$$

FACT 2.12 Any 1-qubit gate can be written as $U(\theta, \iota, \kappa)$ for some $\theta, \iota, \kappa \in [0, 2\pi]$ [cite:need].

DEFINITION 2.13 An **n -qubit gate** is a unitary matrix in $\mathbb{C}^{2^n \times 2^n}$. If gates $U_1, \dots, U_k$ satisfy $U = \bigotimes_{i=1}^k U_i \in \mathbb{C}^{2^n}$, by Fact 2.10.4 U is an n -qubit gate. Special named multi-qubit gates are given in Figure 2.

Name	Circuit	Bra-Ket
CNOT		$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes X$
CCNOT or Toffoli		$ 11\rangle\langle 11 \otimes X + \sum_{\substack{(a,b) \neq (1,1) \\ a,b \in \{0,1\}}} ab\rangle\langle ab \otimes I$
SWAP		$\sum_{i,j \in \{0,1\}} ij\rangle\langle ji $
C-Z		$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes Z$

Figure 2: Common multi-qubit quantum gates

DEFINITION 2.14 Many of the gates in Figure 2 are **controlled** gates. Informally, controlled gates, such as $\text{---}\oplus\text{---}$ only trigger if all of the controls $\text{---}\bullet\text{---}$ connected to it are $|1\rangle$. Formally, for any n -qubit gate U , the single controlled gate, denoted C- U , is defined

$$\text{C-}U := |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U. \quad (12)$$

This definition generalizes for multiple controls. Thus CNOT is just C- X and Toffoli is CC- X . We may suppress the - if the meaning is clear. C- Z is special as it only triggers if both qubits are $|1\rangle$. Note that $I = |0\rangle\langle 0| + |1\rangle\langle 1|$ so $Z = I - 2|1\rangle\langle 1|$. Then

$$\text{C-Z} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes (I - 2|1\rangle\langle 1|) = I \otimes I - 2|11\rangle\langle 11|.$$

Thus, it can be interpreted that both qubits of the C- Z are controls.

2.3 Universality

DEFINITION 2.15 For every $n \in \mathbb{N}$, Let $\text{SU}(n)$ denote the special unitary group under matrix multiplication consisting of $n \times n$ unitary matrices with determinant 1. It is a normal subgroup of the unitary group $U(n)$, consisting of $n \times n$ unitary matrices.

DEFINITION 2.16 A set of gates $\mathcal{S}$ is (efficiently) **universal** if any $U \in \text{SU}(2^n)$, for any $n \in \mathbb{N}$, that can be generated (in $\text{poly}(n)$ gates) by an arbitrary gate set $\mathcal{S}'$, can also be generated by the gate set $\mathcal{S}$ (in $\text{poly}(n)$ gates) [8]. We say a set of gates $\mathcal{S}$ is d -qubit universal if the above holds for $U \in \text{SU}(2^d)$.

REMARK 2.17 In this thesis we will only focus on universality, not efficient universality. Thus, it may take exponential or more gates to simulate certain unitaries U . Furthermore, in practice, the application of U up to some measure and error ϵ suffices. This is called **approximate universality**. This thesis will not focus on approximate universality either.

REMARK 2.18 Definition 2.16 is only concerned with implementing arbitrary unitaries, not necessarily computation. That is, it does not specify whether the inputs and outputs are classical or quantum in nature. This matters because $|\phi\rangle = |0\rangle$ and $|\psi\rangle = -|0\rangle$ have the same measurement distribution and classical output, but $|\phi\rangle \neq |\psi\rangle$. This difference is called **global phase**, i.e. a scalar $e^{i\theta}$ multiplied to an entire state.

DEFINITION 2.19 We use the following types of **universal computation** from [8]:

- **CC-universal**: For every $U \in \text{SU}(n)$, the model can reproduce the distribution of local measurements performed on $U\{|0^n\rangle\}$.
- **QC-universal**: Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in \text{SU}(2^n)$, the model can reproduce the distribution of local measurements performed on $U|\phi\rangle$.
- **CQ-universal**: For every $U \in \text{SU}(2^n)$, the model can produce the quantum state $|\phi\rangle = U|0^n\rangle$.
- **QQ-universal**: Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in \text{SU}(2^n)$, the model can produce the quantum state $U|\phi\rangle$.

FACT 2.20 Any 1-qubit universal gate set and a 2 qubit entangling gate is universal [cite:need].

FACT 2.21 Any universal gate set must include at least one multi-qubit entangling gate, as it is impossible to entangle using only single-qubit gates.

EXAMPLE 2.22 The **Clifford gates** consist of Hadamard H , Phase S , and CNOT. The Gottesman-Knill theorem [7] shows this gate set is not universal and is classically efficiently simulable.

2.4 Measurement-Based Quantum Computation

DEFINITION 2.23 Measurement-based quantum computation (MBQC) roughly proceeds as follows:

1. A classical input specifies the data and program.

2. A resource state $|\phi_k\rangle$ is chosen from some set of states $\Phi = \{|\phi_1\rangle, |\phi_2\rangle, \dots\}$.
3. Local operations (single qubit unitaries), local measurements, and classical communication are used to transform this resource state.
4. By the end, we have a state $|\psi\rangle \otimes |\text{junk}\rangle$. This $|\psi\rangle$ or its measurement is the desired result.

The rest of this section formalizes these notions.

DEFINITION 2.24 A resource state is a quantum state $|\phi\rangle$ that can be used as the initial state in MBQC. A **graph state** $|G\rangle$ is a particular type of resource state associated with an undirected graph $G = (V, E)$ such that:

- For every $v \in V(G)$, create a new qubit in the $|+\rangle$ state using H .
- For every $e = (u, v) \in E(G)$, apply C-Z on qubits associated with u and v .

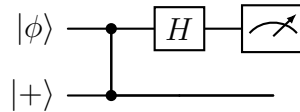
FACT 2.25 There is a unique construction for every graph state, up to reordering of C-Z. This is because C-Zs commute with I and C-Z.

Proof. The C-Z gate is a diagonal matrix, and so is I . The tensor product of diagonal matrices is a diagonal matrix. Furthermore, diagonal matrices commute. Thus the order of adding edges, or equivalently applying C-Zs, does not matter. $\square$

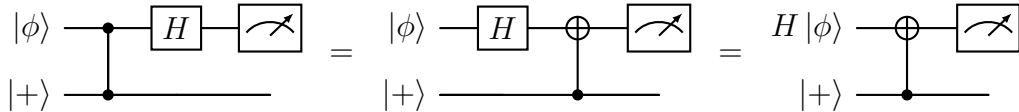
FACT 2.26 Let G be a graph with maximum degree Δ . Then $|G\rangle$ can be constructed in depth $\Delta + 1$.

Proof. Vizing's theorem states that any graph with maximum degree Δ admits an edge coloring with $\Delta + 1$ colors such that no two edges that share a vertex have the same color. All edges of the same color can be implemented in the graph state in the same layer of C-Zs. Thus, at most $\Delta + 1$ layers. $\square$

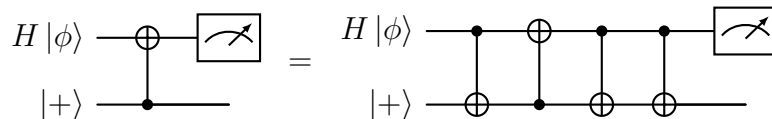
FACT 2.27 Define the Hadamard gadget as the following circuit:



Commuting a Hadamard through the C-Z gives a CNOT:



Two CNOTs cancel each other out and a CNOT has no effect on $|+\rangle$. Then



The first three CNOTs are equivalent to a swap gate. Thus

$$\begin{array}{c}
 H|\phi\rangle \\
 |+\rangle
 \end{array}
 \begin{array}{c}
 \bullet \\
 \oplus
 \end{array}
 \begin{array}{c}
 \oplus \\
 \bullet
 \end{array}
 \begin{array}{c}
 \bullet \\
 \oplus
 \end{array}
 \begin{array}{c}
 \oplus \\
 \bullet
 \end{array}
 \begin{array}{c}
 \boxed{\text{meter}} \\
 \text{---}
 \end{array}
 =
 \begin{array}{c}
 |+\rangle \\
 H|\phi\rangle
 \end{array}
 \begin{array}{c}
 \oplus \\
 \bullet
 \end{array}
 \begin{array}{c}
 \boxed{\text{meter}} \\
 \text{---}
 \end{array}$$

The final state before measurement is

$$\text{CNOT}(|+\rangle \otimes H|\phi\rangle) = \frac{1}{\sqrt{2}}(|0\rangle \otimes H|\phi\rangle + |1\rangle \otimes XH|\phi\rangle).$$

In other words, measuring in the X basis ($\{|0\rangle, |1\rangle\}$) applies an X to $|\phi\rangle$ if we measure a $|1\rangle$. This happens with only 0.5 probability, hence the need for **adaptive measurements**. That is, measurements whose classical outcome can dynamically impact how other qubits are measured.

FACT 2.28 $R_z(\theta)ZR_z(\theta)^\dagger = Z$. That is, $R_z(\theta)$ commutes with C-Z for all θ . Thus,

$$\begin{array}{c}
 |\phi\rangle \\
 |+\rangle
 \end{array}
 \begin{array}{c}
 \bullet \\
 \bullet
 \end{array}
 \begin{array}{c}
 \boxed{R_z(\theta)} \\
 \text{---}
 \end{array}
 \begin{array}{c}
 \boxed{H} \\
 \text{---}
 \end{array}
 \begin{array}{c}
 \boxed{\text{meter}} \\
 \text{---}
 \end{array}
 =
 \begin{array}{c}
 |+\rangle \\
 HR_z\theta|\phi\rangle
 \end{array}
 \begin{array}{c}
 \bullet \\
 \oplus
 \end{array}
 \begin{array}{c}
 \boxed{\text{meter}} \\
 \text{---}
 \end{array}.$$

FACT 2.29 [Euler decomposition] Every single-qubit unitary can be expressed (up to global phase) as the product $R_z(\theta_3)R_x(\theta_2)R_z(\theta_1)$.

FACT 2.30 For all $\theta \in [0, 2\pi]$,

$$R_x(\theta) = HR_z(\theta)H. \quad (13)$$

Thus by Fact 2.29 every single-qubit unitary can be applied to any $|\phi\rangle$ given we are making the right measurements.

THEOREM 2.31 [MBQC on the line is 1-qubit CQ-universal] Suppose you wish to apply gates $U_1, \dots, U_n$ to $|+\rangle$. There is a sequence of single-qubit measurements you can do to the line graph state of $|V| = \Theta(n)$ such that, assuming measurements are all 0, gives $U_1, \dots, U_n|+\rangle$ on the last qubit.

The previous statement is really more of a statement on the computational power (in terms of universality) of the line graph as opposed to MBQC itself. For example, the graph with a single vertex is clearly not universal with the same MBQC gadgets. We formalize this notion below:

THEOREM 2.32 [[8] Observation 2: CQ-universal implies QQ-universal]. A resource is CQ-universal if, using the MBQC protocol (namely, local operations and measurements), we obtain CQ-universality. Every CQ-universal resource can be made QQ-universal by allowing quantum inputs to be coupled into the resource state using the same coupling method as before, i.e. C-Z gates.

2.5 The Grid as a Resource State

DEFINITION 2.33 A $m \times n$ grid graph $G = (V, E)$ has

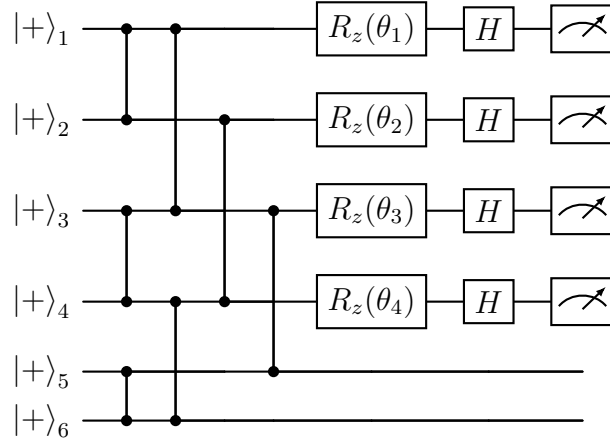
$$V = \{(i, j) \mid i \in [m], j \in [n]\}, \quad (14)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, |i_2 - i_1| + |j_2 - j_1| = 1\} \quad (15)$$

In other words, m rows and n columns with edges between nodes of taxicab distance 1. Denote such graphs as $\text{Grid}_{m \times n}$. Let Grid denote the family of all such graphs.

THEOREM 2.34 Grid is 2-qubit QQ-universal. That is, given a set of 2-qubit gates $U_1, \dots, U_k$ applied to the $|++\rangle$ state, there is a sequence of single qubit measurements you can make on some graph state in Grid , assuming all measurement results are 0, resulting in $U_1 \cdots U_k |++\rangle$ on the last

Proof. By Theorem 2.32, it suffices to show CQ-universality. Consider the following construction:



The above construction can be repeated to apply R_z as many times as necessary. The analysis is similar to Theorem 2.31. That is, it suffices to show that every 2-qubit gate can be decomposed into a sequence of

$$(H \otimes H) (R_z(\theta_1) \otimes R_z(\theta_2)) \text{ C-Z.} \quad (16)$$

The proof of this is essentially brute force [8]. [This thesis' ultimate goal is to recreate this proof, so it should be reproduced here.](#) $\square$

THEOREM 2.35 Grid is QQ-universal.

Proof. [To be done. It is surprisingly challenging to condense.](#) $\square$

DEFINITION 2.36 Let LO denote local operations, i.e. any single qubit unitary. Let CC denote classical communication. Thus MBQC requires LOCC, or both LO through Fact 2.30 and CC due to adaptive measurement.

THEOREM 2.37 [From [8] Observation 5] All efficient universal graph state resources can efficiently be obtained from each other by LOCC.

REMARK 2.38 The previous theorem informally says that the entanglement structure of a resource state is what matters, not its phase. This is because LOCC are precisely the operations that do not add entanglement [cite:need].

2.6 Universality Theorems

DEFINITION 2.39 A family of states Φ is a **universal resource** for MBQC if for all n -qubit state $|\psi\rangle$, $\exists |\phi\rangle \in \Phi$ with $m \geq n$ qubits, such that the transformation $|\phi\rangle \mapsto |\psi\rangle |+\rangle^{m-n}$ is possible with probability 1 with LOCC. We denote this by $|\phi\rangle \geq_{\text{LOCC}} |\psi\rangle$.

THEOREM 2.40 [From [12]] Let Ψ be a family of states and $|\phi\rangle, |\phi'\rangle$ arbitrary states with the same number of qubits. Let $E : |\phi\rangle \rightarrow \mathbb{R}$ be a function that satisfies the following properties:

1. $E(|\phi\rangle) \geq E(|\phi'\rangle)$ whenever $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$;
2. $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$.

Then Ψ cannot be a universal resource.

Proof. The first condition states that LOCC cannot increase this measure, while the second says that the supremum value of E , which can be ∞ , is not reached among all $|\psi\rangle \in \Psi$. If both were true, then (CQ)-universality would be violated. $\square$

REMARK 2.41 E as a measure is so general there are many candidates. However, satisfying the converse of the second condition is not sufficient to guarantee universality. For example, the toric code states have unbounded entanglement width but are not universal [3].

There is more to be said here, but I'm not sure how much of this additional material is necessary, so I've left this out for now.

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